

Ballast worksheet

VBD min counts
VBD max counts
scale mass as flown in field (g)
target thrust for final ballast (g)
target density for final ballast (g/cc)
antenna volume loss (cc)
volmax from regression of field test data (cc)

Deepglider parameters (leave blank for SG/SGX):

abs compress (model or fixed) (m³/dbar)
therm expan (model or fixed) (m³/°C)
ref temperature (°C)
apogee temperature (°C)
apogee pressure (dbar)

comments

Ignore for SG/SGX

Optional box for adding comments
(for record keeping purposes)

Final ballast results

lead to add for final ballast (g)
target mass for final ballast (g)
final scale mass after adjustments (g)
predicted volmax at final ballast (cc)
predicted C_VBD at final ballast
minimum surface density (g/cc)
maximum buoyancy (cc)
neutral stroke (%)

Automatically populated from the most recent log file
← actual mass of glider

Typical values we use for testing in Puget Sound
For open ocean, typical target values are -250 g thrust, 1.0275 g/cc density

← Volume that is out of the water when the glider is at the surface. Antenna is always 150cc.

← volmax - from tank or from regression to field data

← This button calculates the final ballast results from the above parameters

Calculated final ballast and lead to add/remove

← Optional - enter the actual final mass (for record keeping)

← Good starting place for trimming the glider (typically ~200 counts higher than the actual C_VBD if ballasted on ~150m flights for open ocean operating at 1000m)

← The glider will struggle to surface if the surface is less dense than this